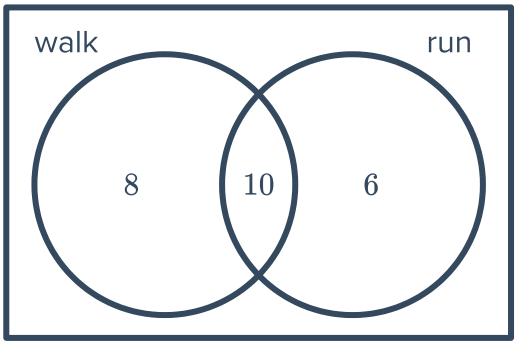
Conditional probability

1 A group of people were asked what type of exercise they like to do. The results are shown in the Venn diagram below:
A person is chosen from the group at ramdom.



a Find the probability that the person walks.

b Find the probability that the person runs, given that they also walk.

2 Two sets of numbers, A and B, are such that set A contains the even numbers from 1 to 20, inclusive, and set B contains the factors of 48 from 1 to 20, inclusive.

a List the numbers in set A.

b List the numbers in set B.

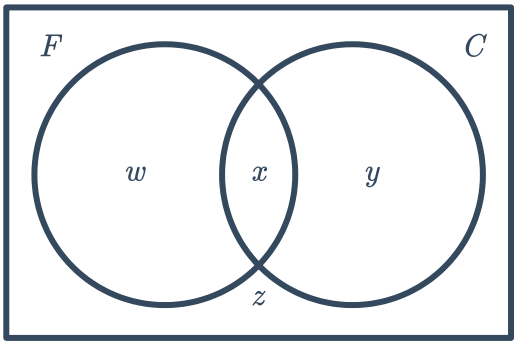
c Find the total number of unique numbers across both sets.

d Find the probability that a randomly selected number is in set B, given that it is in set A.

3 At a university there are 816 students studying first year engineering, 497 of whom are female (set F). Of the 348 students studying Civil Engineering (set C), 237 of them are women.

a Find the value of:

i w **ii** x **iii** y **iv** z



b Find the probability that a randomly selected male student does not study Civil Engineering.

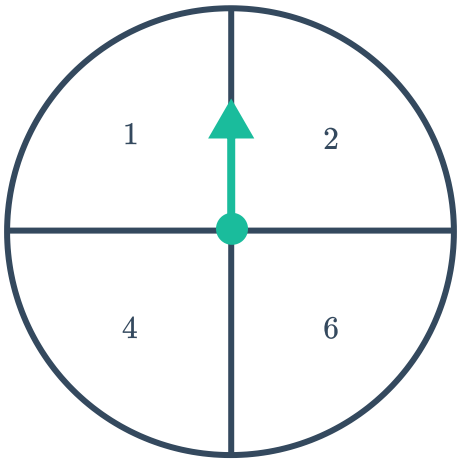
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4 A student is choosing two units to study at university: a language and a science unit. They have 4 languages and 7 science units to choose from.

- **a** If they choose one of each, find the total number of combinations of choices.
- **b** If Italian is one of the languages they can choose from, find the probability they choose Italian as their language.
- **c** French is one of the available languages. Find the probability they choose French as their language given that they choose Chemistry as their science unit.

5 The following spinner is spun and a normal six-sided die is rolled. The product of their respective results is recorded.

- **a** Construct a table to represent all the possible outcomes.
- **b** Find the probability that the product was a multiple of 4 given that a 2 was spun.
- **c** Find the probability that a 6 was rolled given that the product was greater than 10.
- **d** Find the probability that the product was greater than 4 given that the same number appeared on the dice and the spinner.



6 Two cards are randomly drawn without replacement from a deck of cards numbered from 1 to 20. Find the probability that the second card is:

- **a** An even number given that the first card is a 10.
- **b** Less than 5 given that the first one is a 14.
- **c** A number divisible by 5 given that the first card is a 15.

7 Two dice are rolled, one after another. Find the probability, in simplest form, of rolling:

- **a** A pair of fives, given the first die is a five.
- **b** A pair with a sum of 9 or more given that the first die is a four.

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8 Four identical balls labeled 1, 2, 3 and 4 are in a bag. Two balls are randomly drawn from the bag in succession and without replacement. Find the probability, in simplest form, that:

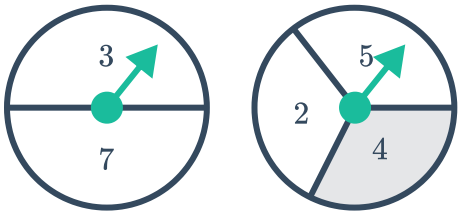
- **a** The first ball is labeled 4 and the second ball is labeled 2.
- **b** The sum of the numbers on the two balls is 5.
- **c** The second ball drawn is 1 given that the sum of the numbers on the two balls is 5.

9 For breakfast each morning, Marge eats porridge, toast or cereal. With that she will either drink orange juice, tea or hot chocolate. Find the probability that Marge will:

- **a** Eat toast and drink tea.
- **b** Eat cereal or drink orange juice.
- **c** Eat porridge given that she drinks hot chocolate.

10 The following two spinners are spun and the sum of their respective spins is recorded:

- **a** Construct a table to represent all possible outcomes.
- **b** Find probability that a 7 was spun given that the sum is less than 12.
- **c** Find the probability that the sum is odd given that a 4 was spun.



11 There are two positions available at a company and the applicants have been shortlisted down to 6: Patricia, Nadia, Amy, Aaron, Lachlan, Jimmy. The two people to fill the position will be picked randomly.

- **Event A:** Patricia is chosen
- **Event B:** Aaron is chosen

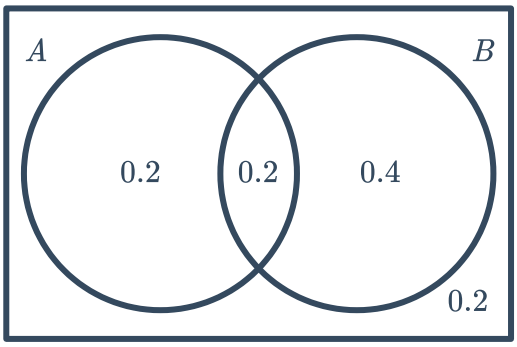
- **a** Describe the probability $P(A|B)$.
- **b** Is $P(A|B)$ less than, greater than or equal to $P(A \cap B)$?

12 Find $P(A \cap B)$ given the following probabilities for an experiment in which A and B are two possible events.

$$P(B|A) = 0.8 \text{ and } P(A) = 0.5$$

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13 Find $P(A|B)$ given the following probability Venn Diagram:



14 In an experiment, there are only 2 possible outcomes, A and B. If outcome A occurs, outcome B does not occur and vice versa. State whether the following statements are true:

a $P(A \cap B) = 1$

b $P(A \cap B) = 0$

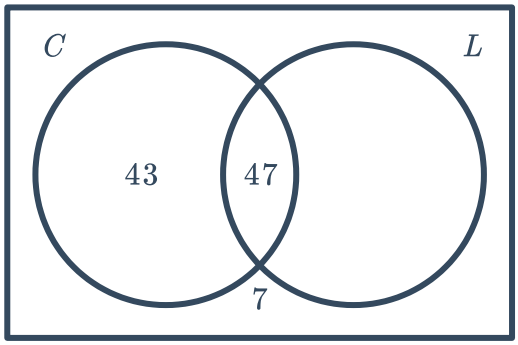
c $P(A \cup B) = 1$

d $P(A) = P(B)$

e $P(A') = P(B)$

f $P(A \cup B) = P(A)$

15 In a survey, 100 office workers are asked about whether they buy a coffee (set C) or lunch (set L) during the work day. Some of the results are shown in the given Venn diagram. Find:



a $n(L \cap C')$

b $P(C|L)$

c $P(L|C')$

16 In a group of 69 people, 45 said they had been to Europe (E), 26 said they had been to Asia (A) and 8 people said they had been to neither Europe nor Asia. Find:

a $P(A \cap E')$ **b** $n(E \cap A)$ **c** $P(E|A)$ **d** $P(A|E')$

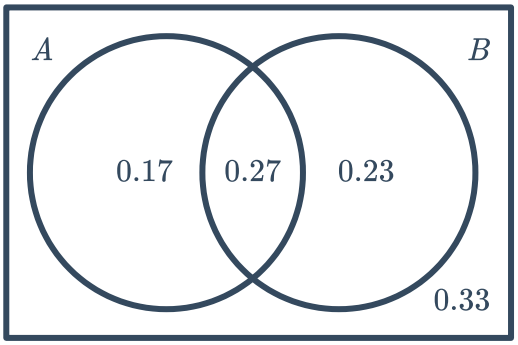
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17 For each of the following Venn diagrams. find:

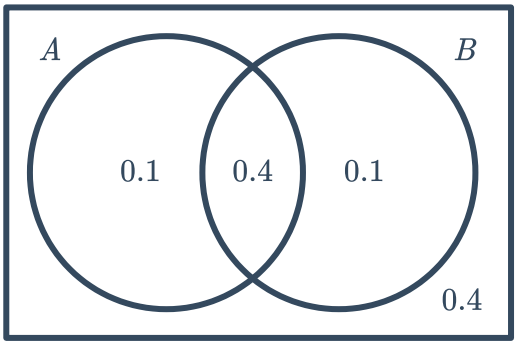
i $P(A' \cap B)$ ii $P(A' \cup B')$ iii $P(A|B)$ iv $P(B|A')$

v $P(A \cup B|B')$

a



b



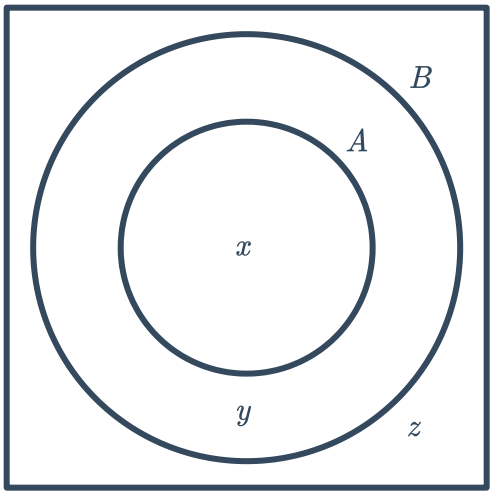
18 Shoppers were surveyed about how much they spend on groceries each week. Of the total surveyed, 426 spent less than $\$200$ each week (set A), 824 spent less than $\$400$ each week (set B) and 154 spent over $\$400$.

a Find the value of:

i x ii y iii z

b Find the probability a shopper spent less than $\$400$ but more than $\$200$.

c Find the probability a shopper spent over $\$400$ if it is known they spent over $\$200$.



19 Consider $P(A) = 0.2$ and $P(B) = 0.5$:

a Find the maximum possible value of $P(A \cup B)$.

b Are events A and B mutually exclusive?

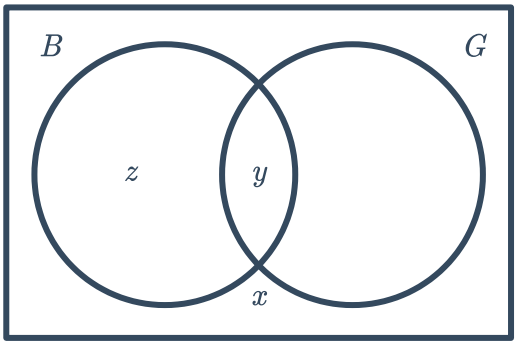
c Find the minimum possible value of $P(A \cup B)$.

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20 In a survey, 1000 people were asked about whether they buy (set B) or grow (set G) their own herbs. The results are listed below:

- 4% said they do neither.
- 40% said they do both.
- Of those who bought their herbs, the probability that they also grew herbs was $\frac{2}{3}$.

This information is to be displayed in the following Venn diagram, where x, y and z represent numbers of people:



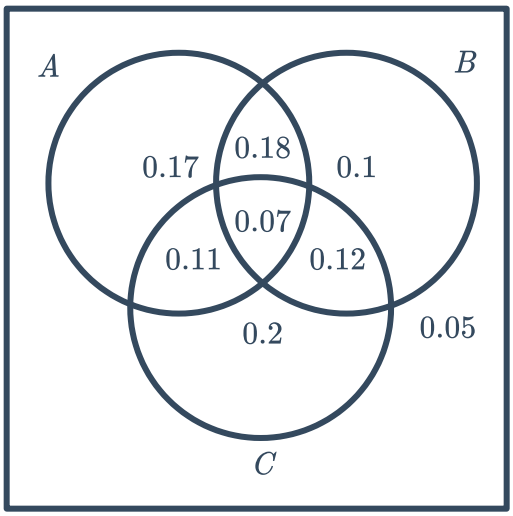
a Find the value of: x and y.

i x **ii** y **iii** z

b Find the number of people who only grew their own herbs.

c Of those who grew herbs, what proportion also bought herbs?

21 Consider the following Venn Diagram:
Find:



a $P\left(A \cap B \cap \overline{C}\right)$

b $P\left((A \cup B) \cap \overline{C}\right)$

c $P\left(B \middle| C\right)$

d $P\left(\overline{A} \middle| C\right)$

e $P\left(A \cup B \middle| \overline{C}\right)$

22 A card is randomly drawn from a standard 52 pack of cards. Find the probability that it's a jack, queen, king or ace if:

a No additional information is known.

b We know it's a 10, jack or queen.

c We know it's a 9 or a queen.

d We know it's not 2, 3, 4 or 5.

23 A basketball team has a probability of 0.8 of winning its first season and 0.15 of winning its first season and its second season. Find the probability of winning the second season, given they won first.

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Independent events

24 Two events A and B are such that $P(A \cap B) = 0.1$ and $P(A) = 0.5$. Calculate $P(B)$ if events A and B are independent.

25 Two events A and B are such that $P(A \cap B) = 0.02$ and $P(A) = 0.2$.

a Are the events A and B are independent if $P(B) = 0.1$?

b Are the events A and B are mutually exclusive?

26 Consider $P(A \cup B) = 0.72$, $P(A) = 0.6$ and $P(B) = 0.3$.

a Find $P(A \cap B')$.

b Find $P(B \cap A')$.

c Find $P(A \cap B)$.

d Find $P(A) \times P(B)$.

e Are A and B are independent?

27 Consider $P(A \cup B) = 0.72$, $P(A \cap B) = 0.18$ and $P(B|A) = 0.6$.

a Find $P(A)$.

b Find $P\left(A \cap \overline{B}\right)$.

c Find $P\left(B \cap \overline{A}\right)$.

d Find $P(A) \times P(B)$.

e Are A and B are independent?

28 Consider $P(A|B) = 0.8$ and $P(A \cup B)' = 0.15$, where A and B are independent:

a Find $P(A)$.

b Find $P(A' \cap B)$.

c Given that $P(A \cap B) = x$, write an expression for $P(B)$ in terms of x.

d Find the value of x.

e State the value of $P(B)$.

29 A and B are two random events with the following probabilities:
Find the value(s) of x if:

- $P(A) = 0.3 + x$
- $P(B) = 0.2 + x$
- $P(A \cap B) = x$

a A and B are mutually exclusive.

b A and B are independent.

c $P(A|B) = 0.6$

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Applications

30 In a survey, 531 people are asked whether they watch My Kitchen Rules (MKR) or Masterchef (MC). Of the total respondents, 177 people watch both and 65 watch neither. The number who watch MKR is twice the number who watch both.

a Find the number of people who only watch MKR.

b Of the people who watch MC, find the proportion that also watch MKR.

c Of those who don't watch MC, find the proportion that watch neither.

31 The following table describes the departures of trains out of a train station for the months of March and April:

	Departed on time	Delayed
March	148	38
April	140	20

a How many trains departed during March and April?

b What percentage of the trains in April were delayed? Round your answer to one decimal place.

c What fraction of the total number of trains during the 2 months were ones that departed on time in March?

d Find the probability that a train selected at random in April would have departed on time.

e Find the probability that a train selected at random from the 2 months was delayed.

32 The following table has been drawn up to show the results of a survey on smoking:

	Men	Women
Smokers	46	96
Nonsmokers	52	100

a How many men were surveyed?

b How many of the people surveyed were smokers?

c What percentage of women were nonsmokers? Round your answer to two decimal places.

d What percentage of nonsmokers were women? Round your answer to two decimal places.

e If a person is selected at random from the group, find the probability that he is a male smoker. Write your answer as a percentage to two decimal places.

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33 Buzz surveyed all the students in Grade 12 at his school and summarized the results in the following table:

	Play net ball	Do not play netball	Total
Height ≥ 170 cm	34	60	94
Height < 170 cm	20	40	60
Total	54	100	154

a What percentage of Grade 12 students whose heights are less than 170 cm play netball? Round your answer to two decimal places.

b A netball player was selected at random from Grade 12. Find the probability that this person is less than 170 cm tall.

c What fraction of the students from Grade 12 do not play netball?

d Find the probability that a student selected at random from Grade 12 is at least 170 cm tall.

34 In a recent study of automobile accidents, the number of passengers and whether the car rolled over was collected:

	Less than 5	5 to 9	10 to 15	More than 15	Total
Roll over	235	15	7	5	262
No roll over	1595	36	50	9	1690
Total	1830	51	57	14	1952

Find the probability that:

a The car rolled over.

b There were 5 or more passengers.

c Given the accident involved less than 10 passengers, the car rolled over.

d Given that the car did not roll over, it had 15 or less passengers.

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35 In a survey, 74 people were asked their age and how many hours they watch television each week. The results are displayed in the following table:

Time spent watching TV per week	< 10 years	10 to 20 years	21 to 30 years	> 31 years	Total
< 3 hours	6	8	2	5	21
4 to 8 hours	10	5	2	4	21
9 to 15 hours	2	10	3	5	20
over 15 hours	3	3	2	4	12
Total	21	26	9	18	74

A person is picked at random from the group. Find the probability that:

a The person is aged less than 21.

b The person watches 3 hours or more of TV a week.

c The person watches less than 9 hours of tv given they are aged over 31.

d A person picked is aged less than 21 given they watch less than 9 hours a week of TV.

36 A test is available to determine whether someone has a particular condition. However it sometimes gives false readings. A false positive is when the person does not have the condition but the test comes back positive, and a false negative is when the person does have the condition but the test comes back negative. A sample of people and their results is presented in the table:

	Have condition	Don't have condition	Total
Test positive	992	980	1972
Test negative	2	9556	9558
Total	994	10 536	11 530

Find the probability that:

a The test is a false positive.

b The test is correct.

c Someone tests positive given that they have the condition.

d Someone who tests positive has the condition.

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37 In a survey, 200 people were questioned about whether they voted for Democrat, Republican or Independent last election and who they'll vote for next election. The rows show who people voted for last election and the columns shows who they'll vote for next election:

	Democrat (next election)	Republican (next election)	Independent (next election)	Total
Democrat (last election)	26	41		70
Republican (last election)			1	90
Independent (last election)	9			
Total		140	10	200

a Complete the table.

b Find the probability that a randomly selected person will vote Democrat next election given that they voted Republican last election.

c Find the probability that a randomly selected person did not vote Independent last election given that they will vote Independent next election.

d If a person voted Democrat or Republican last election, find the probability they'll vote Republican next election.

38 Of the 100 students starting kindergarten surveyed, 80% believe in Santa, 41% believe in the Tooth Fairy and 31% believe in both.

a Complete the following table:

	Believe in Santa	Do not believe in Santa	Total
Believe in Tooth Fairy			
Do not believe in Tooth Fairy			
Total			100

b Of those who believe in Santa, what percentage also believe in the tooth fairy? Round your answer to two decimal places.

c Of those who believe in either Santa or the Tooth Fairy, what fraction believe in both?

d What proportion of those who believe in the Tooth Fairy do not believe in Santa?

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39 In a survey, 200 people were questioned as to whether they read novels in paperback form or on an e-reader:

- 30% said they do both.
- 40% said they read paperbacks.
- The probability that a person who didn't read paperbacks also didn't read on an e-reader was 30%.

a Complete the two-way table for the number of people in each category:

	E-Reader	Not E-Reader	Total
Paperback	60		
Not Paperback			
Total			

b Find the probability that a randomly selected person reads novels on an e-reader but not in paperback.

c Of those that read on e-readers, what percentage read in both formats? Round your answer to the nearest percent.

40 In a survey, 166 children were asked whether they had read or watched at least one of the stories in the Harry Potter series:

- 62 hadn't read any of the books
- 39 had read and watched at least one of the stories
- 97 had watched the movie
- 4 had never read or watched any of these

a Complete the following table for these results:

	Watched the movie	Didn't watch the movie	Total
Read the book	39		97
Didn't read the book		4	
Total		62	166

b Find the probability that a randomly selected student had:

i Read the book but hadn't watched the movie.

ii Read the book given that they had watched the movie.

iii Seen the movie given that they hadn't read the book.

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41 In a population, 15% of people have brown eyes, 25% of people have blonde hair, and 10% of people have both brown eyes and blonde hair. A person is chosen randomly from the population. Find the probability that they have:

a Blonde hair, given that they have brown eyes.

b Brown eyes, given that they have blonde hair.

42 A flight departs from Chicago to Seattle. The following probabilities were found with regards to departure times and weather:

- The probability that the flight departs on time, given the weather is fine in Chicago is 0.8.
- The probability that the flight departs on time, given the weather is not fine in Chicago is 0.6.
- The probability that the weather is fine on any particular day in July is 0.3.

Find the probability that:

a The flight from Chicago to Seattle departs on time in a day in July.

b The weather is fine in Chicago given that the flight departs on time on a day in July.

43 In a survey, 87 people are questioned about whether they own a tablet (T) or a smartphone (S). The following probabilities were determined from the results:

$$P(T|S) = \frac{5}{12}, \quad P(S \cap T') = \frac{35}{87}, \quad P(T) = \frac{14}{29}$$

a Find $n(S \cap T)$. **b** Find $P(S' \cap T)$. **c** Find $P(S|T)$. **d** Find $P(T|S')$.

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